

Performance Testing

Date	29 September 2026
Team ID	SWTID-2026-2508
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing and Spike Testing
Target Module / API	Laptop request catalog item: request submission and approval flow trigger
Test Environment	Cloud-hosted ServiceNow instance
Test Date	29 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal load: employees submit standard laptop requests	10	300	All requests processed within target response time
2	Peak load: multiple requests submitted at the same time	50	300	Response time and error rate stay within targets
3	Approval flow: requests trigger approval and purchase order creation	25	300	Approvals and orders created without failures
4	Spike: sudden surge of requests	100	120	System stays available and recovers after the spike

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Within target under peak load
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Highest value seen during the spike test
3	Throughput (Req/sec)	> 10 req/sec	12.4 req/sec	Pass	Steady throughput during peak load
4	Error Rate	< 1%	0.4%	Pass	Few timeouts during the spike test
5	CPU Utilization	< 80%	62%	Pass	Peak value during the spike test
6	Memory Utilization	< 80%	58%	Pass	Stable throughout the tests

Step 4: Observations & Analysis

Key Findings:

The system met all response time, throughput, error rate and resource targets under normal and peak load, and stayed available during the spike test.

Bottlenecks Identified:

Response time rose slightly when many requests triggered approvals and purchase orders at the same time.

Optimization Steps Taken:

Removed duplicate validation checks and simplified queries in the approval flow, which reduced average response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]